

Formulas

- $RF = \frac{\text{distance travelled by the component}}{\text{distance traveled by the solvent}}$
- Relative atomic mass (A_r) = $\frac{(\text{mass of isotope 1} \times \% \text{ of isotope 1}) + (\text{mass of isotope 2} \times \% \text{ of isotope 2})}{100}$
- Mass = moles $\times$ M_r
- Percentage Yield = $\frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100\%$
- Moles = Concentration $\times$ volume
- $q = m \times c \times \Delta T$
 - o q is the heat transferred, J
 - o m is the mass of water, g
 - o c is the specific heat capacity, $\text{Jg}^{-1}\text{K}^{-1}$
 - o T is the temperature, K
- molar enthalpy change = heat change for the reaction $\div$ number of moles, $\Delta H = q \div n$
- General formula :
 - o Alkane – $C_nH_{(2n+2)}$
 - o Alkene – C_nH_{2n}
 - o Alcohol - $C_nH_{2n+1}OH$
 - o Carboxylic acid - $C_nH_{2n+1}COOH$

Avogadro constant = 6.02×10^{23}
 RTP of gas = 24dm^3 or 24000cm^3

ACRONYMS

- OILRIG - (O)xidation (I)s (L)oss of electron, (R)eduction (I)s (G)ain of electrons
- PANC - (P)ositive (A)node, (N)egative (C)athode
- Reactivity series:

potassium	please
Sodium	send
Lithium	ion
Calcium	Cat
Magnesium	Monkey
Aluminum	and
Carbon	cute
Zinc	zebra
Iron	in
Hydrogen	hot
Copper	countries
Silver	Signed
Gold	Gordon

HALOGENS

- Fluorine - Yellow gas
- Chlorine - Green gas
- Bromine - Red-Brown liquid
- Iodine - Grey solid, purple vapor

% OF GASES

- Nitrogen 78%
- Oxygen 21%
- Argon 0.9%
- Carbon Dioxide 0.04%
- Others 0.06%

Soluble Salts

- All sodium
- All potassium
- All ammonium
- All nitrates
- All chloride except - silver, lead
- All sulfates except - lead, barium, calcium

Insoluble Salts

- All carbonates except - sodium, potassium, ammonium,
- All hydroxides except - sodium, potassium, (calcium)

INDICATORS

Indicators	Colour in acid	Colour in alkaline
Methyl Orange	red	yellow
Phenolphthalein	colourless	pink
litmus	red	blue

- universal indicator
 - red = strongly acidic (pH 0-3)
 - Orange = weakly acidic (pH 4-6)
 - Green = neutral (pH 7)
 - Blue = weakly alkaline (pH 8-10)
 - Purple = strongly alkaline (pH 11-14)

USES AND PROPERTIES

- Aluminum
 - aero planes (low-density)
 - Electricity cables (good conductor, ductile)
 - Saucepans (good conductors, malleable)
- Low-Carbon steel
 - Contains 0.25% carbon
 - hard, strong, malleable, ductile
 - uses: car bodies, bridges, shipbuilding
 - disadvantages: high density(heavy), rust easily
- High-Carbon steel
 - contains 0.6 - 1.2% carbon
 - harder and more resistant to wearing the low-carbon
 - brittle
 - uses: cutting tools e.g. knives, sword
- Stainless steel
 - Contains iron, chromium & nickel
 - Oxide layer prevents corrosion
 - Uses: sinks, saucepans, cutlery, gardening tools

CHEMICAL TEST

GASES

- Hydrogen – lighted splint pops
- Oxygen – Glowing splint relight
- Carbon Dioxide – limewater turns cloudy
- Chlorine – bleaches damp blue litmus paper
- Ammonia – damp red litmus paper turns blue

FLAME TEST

- Clean nichrome wire with HCl and dip it in the sample, hold nichrome wire in blue Bunsen flame
 - (I). Li^+ - Red
 - (II). Na^+ - Yellow
 - (III). K^+ - Lilac
 - (IV). Ca^{2+} - Orange-red
 - (V). Cu^{2+} - Blue-Green
 - (VI). Mg^{2+} - White

TEST FOR CATIONS

- Add sodium hydroxide and test the precipitate formed
 - (I). Ammonium – ammonia gas turn red litmus blue
 - (II). Copper (II) – light blue precipitate formed
 - (III). Iron (II) – Green precipitate formed
 - (IV). Iron (III) – Red-brown precipitate formed

TEST FOR ANIONS

- Carbonate (CO_3^{2-}) – add dilute acid, test gas released - produce CO_2 , turn limewater milky
- Acidify with dilute nitric acid to remove ions & add aqueous silver nitrate - white precipitate formed
 - (I). Chloride (Cl^-) – white precipitate formed
 - (II). Bromide (Br^-) – cream precipitate formed
 - (III). Iodide (I^-) – Yellow Precipitate formed
- Sulfate (SO_4^{2-}) – Acidify with dilute nitric acid & add aqueous barium Nitrate - White precipitate formed

UNSATURATED OR SATURATED HYDROCARBONS

- Add Bromine water/solution
 - Saturated – no change without UV
 - Unsaturated – changes colour from orange to colourless

REACTIONS

- Metal + Acid $\rightarrow$ Salt + Hydrogen
- Metal Hydroxide + Acid $\rightarrow$ Salt + Water
- Metal Oxide + Acid $\rightarrow$ Salt + Water
- Metal Carbonate + Acid $\rightarrow$ Salt + Water + Carbon dioxide

Types of reaction

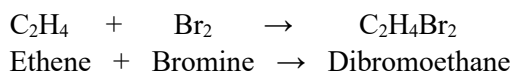
- **Substitution**

- A substitution reaction takes place when one functional group is replaced by another
- Example: Methane reacts with bromine under ultraviolet light



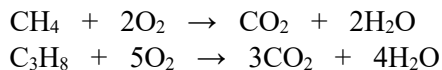
- **Addition**

- An addition reaction takes place when two or more molecules combine to form a larger molecule with no other products
- Example: Bromine will react with ethene and the bromine molecule will react and add across the double bond of the ethene

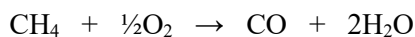


- **Combustion**

- This is the scientific term for burning. In a combustion reaction, an organic substance reacts with oxygen to form carbon dioxide (or carbon monoxide if incomplete combustion) and water.
- Example: Alkenes burn when heated in the air of oxygen
- If there is an unlimited supply of air/oxygen, the products are carbon dioxide and water:



- This is termed complete combustion
- If there is a limited supply of air/oxygen, the products are carbon monoxide and water:



- This is termed incomplete combustion

SEPARATION OF MIXTURES

- Crystallisation (used to separate self-produced soluble salt from the solution)
 - React
 - Filter to remove excess reactant
 - Heat to evaporate the solvent
 - Test if the solution is saturated by dipping glass rod, crystal would form on the glass (if saturated)
 - Cool & filter the solid crystal that form
 - Wash the crystal with cold distilled water
 - allow them to dry in a warm place
- Simple distillation (used to separate a liquid and soluble solid from a solution)
 - heat the solution in a boiling flask which connect to a condenser
 - pure liquid evaporates
 - vapours cool down in the condenser
 - Collect pure liquid from the condenser in a beaker
 - After all the liquid has evaporated only solid solute is left behind
- Fractional distillation (used to separate a mixture of two or more liquids with different temperatures)
 - (*Liquid A – Lowest boiling point, Liquid B – Second lowest boiling point*)
 - Heat up to the boiling point of the liquid A
 - Liquid A will evaporate first
 - pass the vapours in a condenser to cool & condense
 - Pure liquid A will form in the condenser
 - Collect pure liquid A in a beaker
 - Stop heating immediately when the temperature starts reaching the boiling point of liquid B
- Filtration (used to separate undissolved solids in a liquid)
 - Place filter funnel with filter paper on it above a beaker
 - Pour the mixture into the filter funnel
 - Collect liquid in the beaker
 - Undissolved solids should be left behind in the filter funnel
- Fractional distillation(used to separate crude oil)
 - Crude oil is heated so vapours rise
 - Vapours of hydrocarbons with very high boiling points will immediately condense into liquid at the higher temperatures lower down and are tapped off at the bottom of the column
 - Vapours of hydrocarbons with low boiling points will rise up the column and condense at the top to be tapped off
 - The different fractions condense at different heights according to their boiling points and are tapped off as liquids
 - The fractions containing smaller hydrocarbons are collected at the top of the fractionating column as gases
 - The fractions containing bigger hydrocarbons are collected at the lower sections of the fractionating column

- Paper Chromatography (used to separate substances that have different solubilities in a given solvent)
 - Draw a pencil line on chromatography paper
 - Place spots of the sample
 - Lower the paper into a solvent container of hot water
 - Make sure the pencil line sits above the level of solvent
 - Solvent travels up the paper taking along some substance with it
 - Different substance with different solubilities travels at different rates
 - Which shows the different components the sample contains

MAKING SALTS

Soluble

- (I). Crystallisation-
 - React (except NH_4^+ , K^+ , Na^+) : acid + Metal oxide/hydroxide/carbonate
 - Filter to remove excess reactant
 - heat to evaporate water
 - Cool & collect crystals that form
 - allow them to dry in a warm place
- (II). Titration:
 - React (NH_4^+ , K^+ , Na^+)
 - Acid (in a burette) + Metal hydroxide/carbonate (in a conical flask)
 - Acid (in a burette) + ammonia solution (in a conical flask)
 - add an indicator to know when the alkali is neutralized
 - add acid from the burette until the indicator changes colour and record acid volume
 - Repeat with the same volume of alkali and acid without any indicator
 - heat to Evaporate
 - Cool and collect crystals that form
 - allow them to dry in a warm place

Insoluble

- (I). Precipitation
 - React : Soluble salt₁ + Soluble salt₂ → Insoluble salt + Soluble salt₃
 - Cool the solution
 - Filter the Insoluble salt from soluble salt₃ solution
 - Wash the insoluble salt
 - allow them to dry in a warm place

Appearance

- Magnesium Oxide – White powder
- Magnesium ribbon – Shiny/Silvery
- Silver chloride – White
- Copper Oxide – Black
- Copper(II)Carbonate – Green
- Bromine – Red-brown liquid

Naming Organic Compound

First Part of the name		Second part of the name		
Name	Number of carbon atoms	Name	Functional Group	Family
Meth...	1	...Ane	None	Alkane
Eth...	2	...Ene	C=C	Alkene
Prop...	3	...Anol	R-OH	Alcohol
But...	4	...Anoic Acid	R-C(=O)-OH	Carboxylic Acid
Pent...	5	...Amine	R-NH ₂	Amine
Hex...	6	...Yl...Anoate	R-C(=O)-O-R	Esters